Kairos: from observed history to executable progress

A single ready order, a separate capacity transaction, and layout-aware state continuity

Causal state snapshot

Arrived prompt | Completed history | Actual KV layout | Ready / running / tool

1 Information

Two forests
48 trees each

Current output
P(O | history)

Post-current service
P(R+ | history)

Predict on arrival; update after completion.
Use a suffix stage prior while pending.

2 Program scheduling

J = physical cost / expected progress

Ready order: **A** **B** **C** **D**

Increasing J; cold and returning calls together

Independent capacity transaction

Native-frontier pages + execution slots
Earlier claim + eligible optional victims

4 Restore & execute

GPU pages

CPU chunks

Use the joint recoverable prefix.
Copy missing pages; recompute gaps.

Native allocator + vLLM / LMCache

• Program & tool

LLM → tool wait → next arrived call

3 KV continuity

Keep benefit: actual recovery saved

Return probability × benefit − pressure × hold time

Feedback contains completed observations only. Solid: decision / execution flow. Dashed: completed-state feedback.